

Paper B-05: Neural Computation: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Abstract

This paper treats neural computation as an engineered and biological transport analogy for CDFD/AFL. Spikes, synaptic current, activation flow, gradients, and attention-like routing are read as candidate fluxes Φ moving through constraints C such as bandwidth, inhibition, metabolic cost, finite precision, and connectivity. The paper does not claim that biological brains and artificial networks are the same material system. It asks which measurable bottlenecks and memory terms make neural computation auditable.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow-constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine [5]. Part IV [6], DOI <https://doi.org/10.5281/zenodo.20395901>, supplies

the cross-domain stress-test archive. This paper sits in Part B because it focuses on the engineered-computation reading of neural systems.

2 Neural Computation as Transport

Neural computation moves signals through constrained pathways. In biological tissue, those constraints include synaptic strength, inhibition, refractory periods, metabolic cost, and network topology. In artificial neural networks, they include parameter count, activation bandwidth, context length, memory traffic, precision, optimization state, and hardware limits. CDFD/AFL gives a single audit question for both settings: where does flow concentrate, what limits it, how does the system adapt, and what past load remains in the structure?

3 Candidate Variable Map

CDFD term	Neural-computation reading	Example proxy
Φ	signal, activation, or gradient flow	spike rate, activation norm, token updates
C	active constraint	inhibition, bandwidth, precision, memory traffic
S	responsiveness	plasticity, gating, normalization, routing adjustment
M_s	structural memory	synaptic weights, trained parameters, cached state
Ψ_s	operating ratio	balanced, saturated, unstable, or underused regime

4 Biological and Engineered Boundaries

Part III gives the biological and medical context for AFL [5]. This paper uses that context carefully. A biological neuron, a neuromorphic device, and a transformer layer can all be described as constrained signal systems, but the measurements differ. A useful comparison must state the unit of flow, the time scale, the constraint, and the failure mode before drawing any cross-domain conclusion.

5 Measurement Layer

For biological data, measurable proxies may include firing rate, synaptic current, local field power, metabolic demand, refractory recovery, and plastic change. For artificial systems, proxies may include activation magnitude, gradient norm, attention entropy, memory bandwidth, context pressure, and inference latency. CDFD/AFL adds value only if these proxies improve the audit of saturation, collapse, recovery, or adaptation.

6 Current Runtime Diagnostic

The Part B adapter sweep includes artificial-intelligence and computation rows, and the Part E cascade test records finite runtime behavior in the current CDFD Runtime [7]. These files document model behavior. They do not validate a neuroscience claim or certify an AI-system claim.

7 Tests That Can Break the Mapping

The mapping weakens if measured bottlenecks do not track saturation or recovery, if ordinary neural-network or neuroscience metrics explain the behavior more cleanly, or if the proposed variables cannot be measured without circular definitions.

8 Major Universal Upgrade: Mechanism, Scale, and Tests

8.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Neural Computation, the proposed drive is signals, training examples, gradients, and inference demand. The active limitation is compute, data quality, architecture, bandwidth, and noise. Responsiveness is carried by learning, attention, sparsity, regularization, and plasticity, while retained state is represented by weights, optimizer state, replay history, and accumulated bias. The outcome to explain is accuracy, calibration, generalization, and adaptation.

Table 1: Operational CDFD/AFL map for Paper B-05.

Term	Paper-specific observable meaning	Measurement question
Φ	signals, training examples, gradients, and inference demand	What rate, load, gradient, or demand enters the focal system?
C	compute, data quality, architecture, bandwidth, and noise	Which measured bottleneck limits transfer, function, or recovery?
S	learning, attention, sparsity, regularization, and plasticity	Which process changes effective capacity or reroutes the load?
M_s	weights, optimizer state, replay history, and accumulated bias	Which retained state makes equal present loads produce different futures?
Y	accuracy, calibration, generalization, and adaptation	Which outcome is predicted out of sample, with uncertainty?

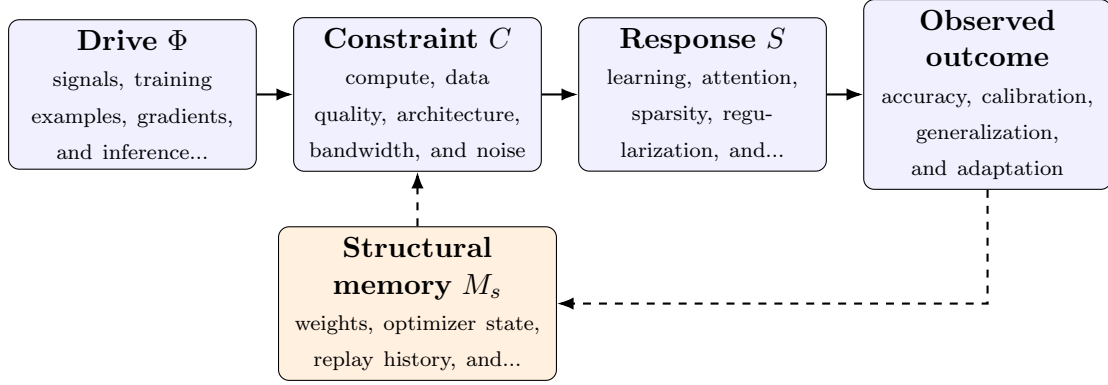


Figure 1: Paper B-05 observable CDFD/AFL architecture for Neural Computation. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

8.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \quad (1)$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \quad (2)$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \quad (3)$$

$$\Psi_s(t) = \frac{\hat{\Phi}(t)}{\epsilon + \hat{C}(t)} \hat{S}(t) [1 + \omega \hat{M}_s(t)]. \quad (4)$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in signals, training examples, gradients, and inference demand arrives faster than learning, attention, sparsity, regularization, and plasticity can compensate. This raises or redistributes compute, data quality, architecture, bandwidth, and noise. If the load persists, the retained state encoded by weights, optimizer state, replay history, and accumulated bias changes the next response, so a later event of equal magnitude can produce a different trajectory in accuracy, calibration, generalization, and adaptation. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

8.3 Cross-Scale Universality Without Mechanistic Erasure

For the engineered systems family, the universal comparison is between demand, designed capacity, feedback control, dependency topology, and accumulated technical state. The

strongest engineering use is prospective: the mapping must identify a bottleneck, a control action, or a recovery signature before the failure occurs. [2, 9, 8, 1] The CDFD programme is cumulative: Part I supplies the flow-constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is nanoseconds to model lifetimes; units to learning ecosystems. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

8.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper B-05. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure learning curves, activation traces, gradient statistics, ablations, and benchmark shifts and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a distribution shift or capacity bottleneck after training while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: explicit constraint-memory variables do not improve generalization or adaptation forecasts.

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ , separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

8.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from learning curves, activation traces, gradient statistics, ablations, and benchmark shifts and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL

state extension and test whether it improves prediction of accuracy, calibration, generalization, and adaptation. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

9 Conclusion

Neural computation is a useful Part B case because it sits between biological signal processing and engineered computation. The CDFD/AFL frame is strongest when it identifies real bottlenecks and memory terms rather than replacing existing domain models.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

10 Reference Layer

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